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Review

Satellite Remote Sensing for Crop Yield Prediction: A Review

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Abstract

The rapid evolution of Earth observation satellite missions and computational methods made satellite remote sensing a foundation of state-of-the-art crop yield prediction. Therefore, the aim of this review is to analyze dominant drivers of crop yield prediction research based on satellite remote sensing, including dominant sensor types, satellite missions, crops, and specific research topics, as well as to identify present issues and research gaps. This review summarizes the bibliometric analysis of satellite-based crop yield prediction publications during 2000–2025, including 1174 articles that were indexed in the Web of Science Core Collection. Annual publication and citation trends, geographic patterns of research publications, prevalent satellite missions and sensor types, predominant crops used in research and trends in research themes were analyzed in the study. Findings show that there has been a consistent expansion of the study topic regarding publication count, with multispectral data, especially that of Sentinel-2, Landsat, and MODIS missions, being utilized in most of the literature in the field, while radar-based approaches are becoming increasingly important, providing complementary data to multispectral imagery. The review indicates a methodological shift in the models of simple regressions to machine learning, deep learning, and multi-sensor data fusion frameworks that use dense satellite imagery time series.

Keywords: bibliometric analysis; vegetation indices; synthetic aperture radar; land surface phenology; machine learning; deep learning; data fusion; precision agriculture

1. Introduction

Accurate and timely crop yield prediction has formed the foundation of food security planning, agricultural policy development, and stabilization of the market regionally and globally [1]. The information on subfield yield potential zones is used to make decisions in the production of arable and permanent crops, including farm-level management and insurance evaluation, national grain reserve planning and international trade forecasting [2]. The conventional techniques of yield prediction that use a combination of field surveys, farmer reports and statistical extrapolation are usually labor-intensive, time consuming and spatially constrained [3]. This restricts the ability to capture certain intra-seasonal variables and respond to abnormal weather phenomena and also restricts the research in terms of spatial coverage, temporal resolution and operational flexibility. These limits are extreme in large agricultural regions, data-sparse environments or in rapidly changing climatic environments [4]. As a result, there has been long-lasting interest in alternative approaches that can produce consistent, scalable and timely information for a variety of agroecological systems. Satellite remote sensing has become one of the most promising

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data sources for overcoming these challenges [5]. Satellite imagery allows constant tracking of crop development and conditions in a large space using easily accessible multitemporal imagery of the Earth's surface [6]. With the early introduction of coarse resolution sensors to agricultural monitoring, development of sensor technology, availability of data and computation methods drastically increased the possibilities of satellite-based crop prediction [7]. The modern satellite missions provide multispectral, hyperspectral, thermal, and radar observations of progressively higher spatial and temporal resolution, enabling monitoring of important biophysical and phenological processes in crop development [8,9].

The main assumption of satellite-based yield prediction is that productivity of crops is related to canopy properties which can be reliably observed [10]. The leaf-area index, chlorophyll content, biomass accumulation and water status were found to affect the spectral and backscatter signals of sensors from various satellite missions [11,12]. Crop vigor and photosynthetic activity have traditionally been assessed using vegetation indices derived using optical data, such as the Normalized Difference Vegetation Index (NDVI) [13] and the Enhanced Vegetation Index (EVI) [14]. On a similar note, synthetic aperture radar (SAR) backscatter measurements have proven to be sensitive to structure, biomass and moisture conditions of crops, which have specific advantages in cloud-prone locations where optical measurements are often not available [15,16]. Through the relationships of these remotely sensed indicators with the measured ground truth yield results, a broad spectrum of empirical, semi-empirical and machine learning-based yield prediction frameworks were developed [17–19]. Initial research using satellite-based remote sensing data in crop yield prediction mainly depended on statistical regression techniques with the single-date or seasonal maximum vegetation indices typically used as covariates [20]. Although these techniques proved that reliable yield prediction was possible, they were often limited in performance by interannual variability, saturation effects and lack of model generalizability related to region or crop specificity. With the increased availability of satellite imagery time series, studies started to take advantage of the temporal aspect of remotely sensed data and combine information during the growing season to obtain a more accurate representation of crop dynamics [21]. The phenological measurements (peak of season and duration of vegetative period) have since been added in yield models because of the transition to the spatiotemporal component of crop development [22]. Simultaneously with the further development of remote sensing, machine learning and data-based model creation have redefined the field of crop yield prediction. Algorithms including random forests, support vectors machines, neural networks, and, more recently, deep learning architectures have become commonly used due to their ability to model nonlinear, complex relationships between yield and multi-source abiotic covariates [23,24]. They are especially applicable to high-dimensional satellite data, in which interactions among crop yield, spectral indices and abiotic properties can be challenging to model with conventional regression [25].

Despite these developments, crop yield prediction based on satellite imagery does not regularly achieve high accuracy of yield variance. Crop yield only partially depends on vegetation properties observable from space, also depending on other attributes like soil properties, management practices, pests and diseases, and extreme weather events [26,27]. These properties can only partially be represented by satellite imagery, or not at all in the case of management practices [28]. Additionally, the importance of remotely sensed covariates on yield can have considerable differences among crops, regions and years, making it more difficult to generalize models [29–31]. Problems associated with data quality such as cloud cover, sensor noise and temporal gaps also impact model robustness, especially when operational prediction is involved [32]. Although a lot of previous research has emphasized regional or country-level yield predictions [33,34], subfield yield prediction is increasingly in demand due to variable rate technology application in precision agriculture [35]. Satel-

lite imagery from sources such as Sentinel-2 and Landsat 8 and 9, as well as commercial missions with high spatial resolution, made subfield studies possible but were also affected by subfield heterogeneity, spectral mixing and accurate field boundary information [36]. There has also been growing interest in the use of multi-source and multi-sensor data in predicting crop yield in recent years [37]. Integration of optical and radar data would be able to increase resistance to cloud cover, particularly for winter crops [38], while incorporation of meteorological data, soil data, and management variables would provide a more detailed representation of crop yield dynamics [39].

With the increasing size of historical satellite imagery archives and the increasing availability of computational resources, research in crop yield prediction rapidly increased in recent years. The growing literature of this type is an indication of a large variety of methodological options, combinations of sensors, spatial resolutions, and assessment approaches. Therefore, the aim of this review is to analyze dominant drivers of crop yield prediction research based on satellite remote sensing, including dominant sensor types, satellite missions, crops, and specific research topics, as well as to identify present issues and research gaps.

2. Materials and Methods

This review is based on bibliometric analysis of publications indexed in the Web of Science Core Collection (WoSCC) using the advanced search query "TS = (crop yield AND (predict* OR forecast*) AND (satellite remote sensing OR satellite imagery))". The query was constructed to outline a specific research scope rather than aiming to include all related aspects of agricultural monitoring and production assessment. The query focused on the publications that merge "crop yield" and predicting terms "predict" and "forecast" with satellite remote sensing, with the intention of selecting studies that use yield prediction as a core methodological focus, rather than mentioning it within the wider contexts of crop monitoring, early warning, or agronomic assessment frameworks.

A total of 1199 studies were obtained from a single export of the WoSCC, and no duplicate records were identified or manually removed as a single database was analyzed. Only studies in the English language classified as articles and proceeding papers were included in the review. The exported data included article titles (TI), abstracts (AB), author keywords (DE), and Keywords Plus (ID). The review period was set to 2000–2025, resulting in the exclusion of 25 studies, resulting in 1174 studies that were analyzed in the review. All calculations in the review were based on a combined text field generated by concatenating titles, abstracts, author keywords, and Keywords Plus and converting all text to lowercase.

Annual publication counts, total citations, and mean citations per article were calculated to characterize temporal trends. Publication counts were derived from author affiliation data (AU_CO). Scientific impact by country was assessed using total citations and the H-index, calculated from article citation counts. Geographic distributions of publications were visualized using world map polygons with log-transformed publication data.

The temporal dynamics of scientific production were analyzed through life cycle analysis based on model logistic growth, which represents a well-known approach to bibliometric research that is employed in describing the emergence, growth, maturation, and saturation stages of a research topic. The number of publications every year was modeled as a logistic growth curve, which supposes that the process of scientific production has an S-shaped appearance, with an initial period of slow growth, a high level of expansion, and an eventual plateau. The logistic model is an estimation of the cumulative publication number in relation to time, and the rate of publications per year is its first derivative. Additionally, Fisher–Pry transformation was used to confirm the logistic growth analysis

of the cumulative publication data [40], including a linearization of the logistic curve that represents growth as a logarithm of the ratio of the cumulative fraction of publications to its complement.

Publication analysis according for major satellite missions was performed by searching for strings "sentinel", "sentinel-1", "sentinel-2" or "sentinel-2" for Sentinel missions; "landsat", "landsat 5", "landsat 7", "landsat 8" or "landsat 9" for Landsat missions; "modis" for Moderate Resolution Imaging Spectroradiometer (MODIS); and "planetscope", "planet", "rapideye", "worldview", "geoeye" and "ikonos" for high-resolution commercial missions. Studies matching two or more satellite mission classes were classified as multi-mission studies.

Sensor types were grouped into multispectral, radar, and hyperspectral classes based on strings "sentinel-2", "landsat", "modis", "planetscope", "rapideye", "worldview", "geoeye" or "ikonos" for multispectral; "sentinel-1", "radarsat" or "terrasar-x" for radar; and "hyperion", "prisma", "enmap" or "aviris" for hyperspectral sensors. Studies matching two or more sensor types were classified as multi-sensor studies.

Major crops were identified by searching for strings "maize" or "corn" for maize; "wheat" for wheat; "rice" for rice; "soybean", "soy bean" or "soya" for soybean; "barley" for barley; "sorghum" for sorghum; "sunflower" for sunflower; "rapeseed" or "canola" for rapeseed; "cotton" for cotton, "potato" for potato; and "vegetable", "tomato", "pepper", "cucumber", "onion", "lettuce", "carrot", "broccoli", "cabbage", "spinach", "eggplant", "aubergine", "zucchini" or "courgette" for the generalized vegetable class. If there were multiple occurrences per study, publications were assigned to multiple crop classes. Particular crops in the same study with coauthors from two or more countries were assigned multiple times to countries for each coauthor according to their entry in the AU_CO attribute exported from the WoSCC database.

3. Present State of Crop Yield Prediction Research Based on Satellite Remote Sensing

3.1. Life Cycle Analysis of Study Topic

Figure 1 represents the development of research on crop yield prediction using satellite remote sensing over the period of 2000–2025. Based on historical records of publications indexed in the WoSCC, Figure 2 includes a life cycle analysis with a projection of future development of this research topic. The formation of the early stage of the research life cycle, which occurred in the 2000s, was characterized by low rates of publication and a low rate of citation. At that time satellite-based yield prediction studies were predominantly in the exploration phase due to limited alternatives in satellite missions, mainly including Landsat and MODIS, as well as limited computational capabilities [41,42]. However, this early stage of development formed the conceptual basis by developing empirical associations between vegetation indices and crop productivity [43,44]. This expansion of publications in the mid to late 2010s is proportional to the broader usage of satellite imagery in crop yield prediction. This coincided with the launch of the Landsat 8, Sentinel-1 and Sentinel-2 satellite missions, providing improved capabilities for multitemporal analysis of crop phenology [45–47]. The growing number of citations in the period suggests that the research started to gain greater scientific applicability and methodological stabilization. The steep increase in the number of publications and cumulative citations in the late 2010s and early 2020s is also a consequence of the integration of more machine learning methods which employed dense time series and high dimensional feature spaces in crop yield prediction [48–50].

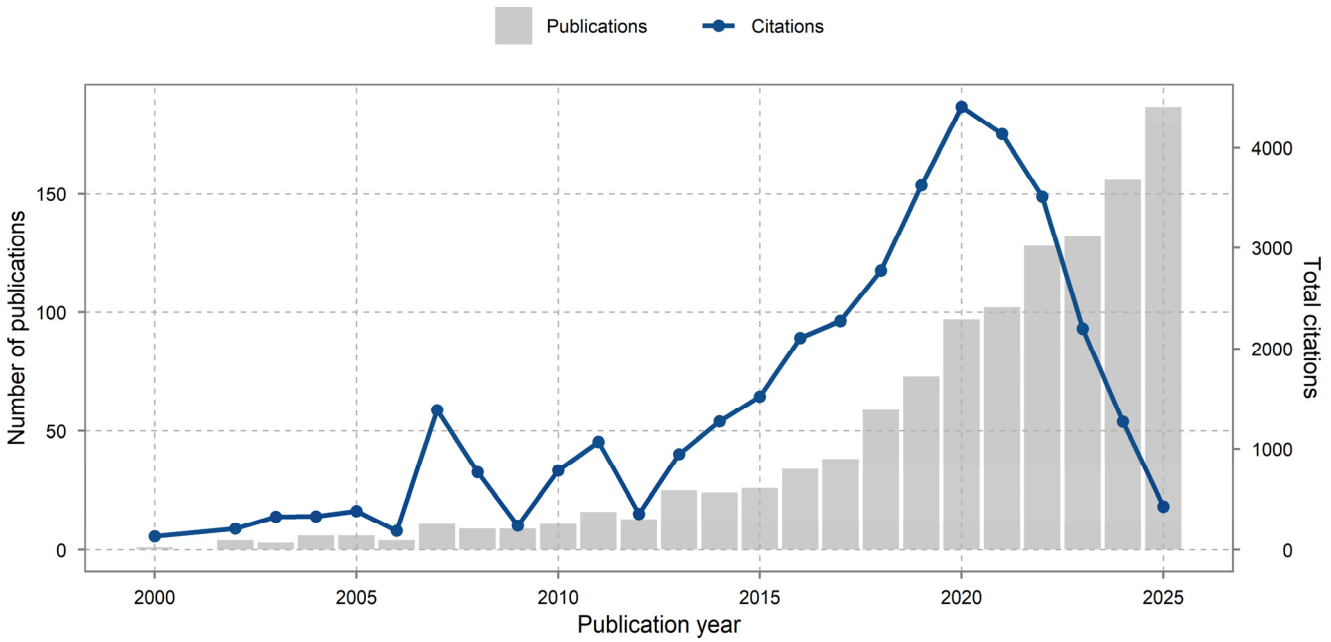


Figure 1. Yearly number of crop yield prediction publications based on satellite remote sensing (gray bars) and total citations (blue line) from the WoSCC database for the 2000–2025 period.

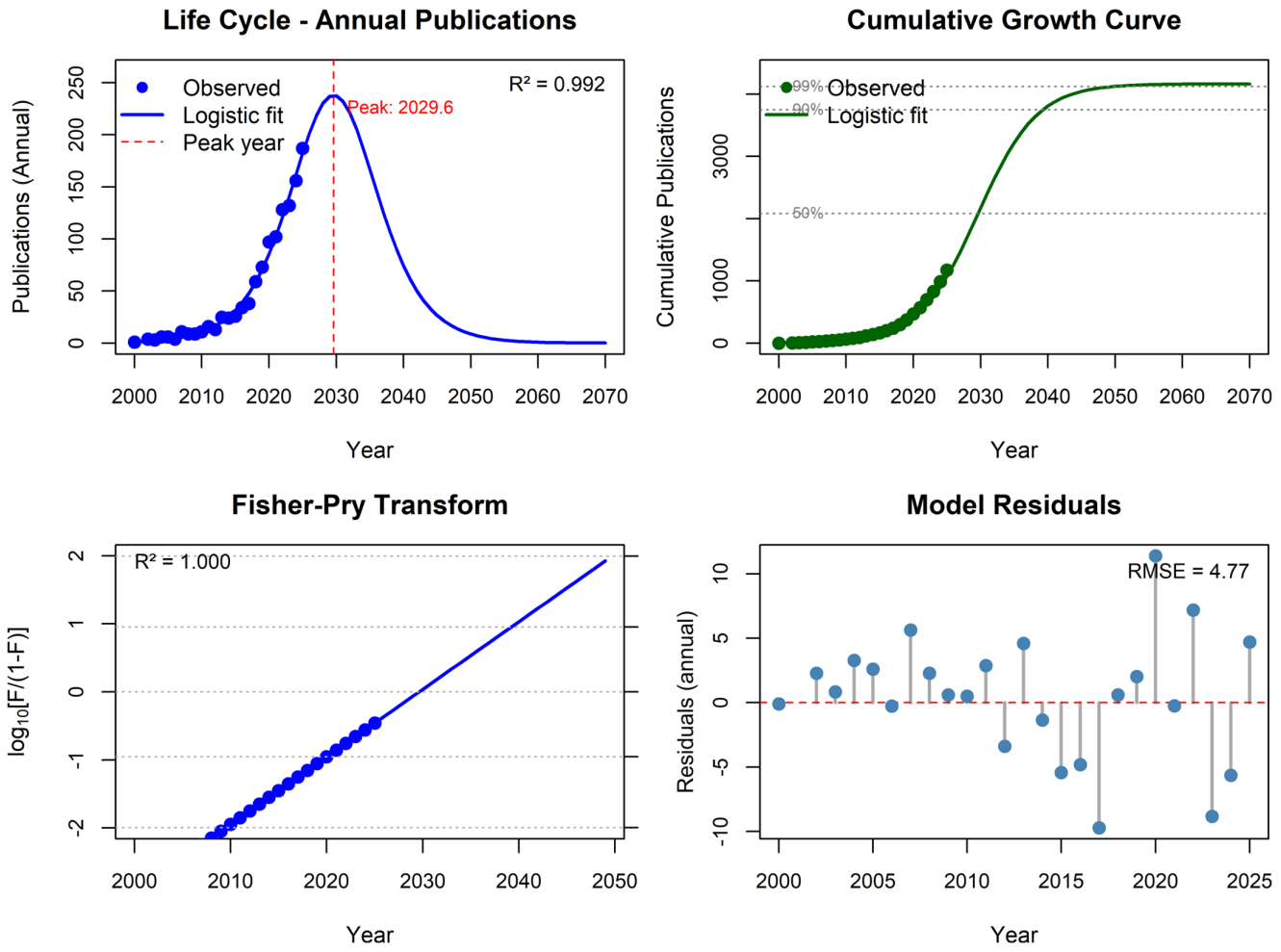


Figure 2. Life—cycle analysis of the study topic according to logistic fit and Fisher–Pry transformation.

However, the period between 2020 and 2025 is characterized by a rapid increase in the number of yearly publications with a severe drop in the number of total citations. Since lag in the number of citations is natural regardless of the life cycle of the study topic [51], citation data in the next few years shall reveal whether it reached its peak or it will continue growing in the same trend. Life cycle analysis based on the logistic fit of yearly publications since the year 2000 suggests that the study topic is nearing its peak with a very high-fitting coefficient of determination, which could be reached before the year 2030. However, the estimated saturation phase by 2030 should be interpreted with caution. While traditional multispectral approaches may be reaching a maturity plateau, the recent emergence of multimodal Large Language Models (LLMs), vision transformers, and next-generation hyperspectral missions (e.g., CHIME, EMIT) could catalyze a new growth cycle, effectively disrupting the projected logistic plateau [52,53]. Additionally, the results of Fisher–Pry transformation strongly suggest that the dynamics of publications are in the logistic growth form and validate the estimated life cycle parameters. The absence of a strong temporal trend in calculated residuals indicates that there is no systematic bias in the model fit, as the root mean square error (RMSE) indicates low average deviation of the observed and predicted values. The logistic model suggests a strong fit, but increasingly further projections in the future beyond the current data increase the uncertainty of the predictions. The non-zero residuals suggest that the yearly variation in publication activity is the result of extrinsic influences, such as funding cycles, global events and even new interdisciplinary connections. This suggests that the used logistic fit is a possible scenario in the future but is subject to many potential disruptions. Moreover, the increase in the cumulative influence of a field may be the result of a small number of publications that have received a disproportionately high number of citations, which adds uncertainty to the analysis of the trends in potential future knowledge growth.

3.2. Dominant Satellite Missions and Sensor Types for Crop Yield Prediction

Figures 3 and 4 display a representation of dominant satellite missions and sensor types in crop yield prediction studies based on satellite imagery, respectively. The distribution of indexed publications strongly indicates the primary importance of multispectral optical sensors, especially Sentinel-2, Landsat and MODIS, which were included in the greatest number of the publications. Multispectral sensors are dominantly represented as optical data and were used to calculate vegetation indices to measure crop vigor, biomass accumulation and photosynthetic activity, which were used as proxies for crop yield [54]. Among these missions, Sentinel-2 became increasingly dominant since the year 2020 due to its high spatial resolution, short revisit time, and red-edge bands, which are particularly sensitive to the biophysical crop properties [55]. The extensive historical record of Landsat imagery is a major component of its relevance, which makes it possible to model the yield of multi-decade trends [56]. An increasing presence in research is also demonstrated by radar sensors, especially Sentinel-1, which complement multispectral imagery in crop monitoring due to their resistance to cloud cover, which is especially relevant for winter crop yield prediction [57,58]. Radar backscatter also provides information on crop structure, surface roughness and moisture conditions that cannot be directly represented through multispectral imagery [59]. Although they presently have a lower share of publications, hyperspectral sensors are an indication of a new field of research. However, present limitations in the data availability due to costs and processing complexity limit their applicability relative to open data satellite missions, such as those of Sentinel and Landsat [60].

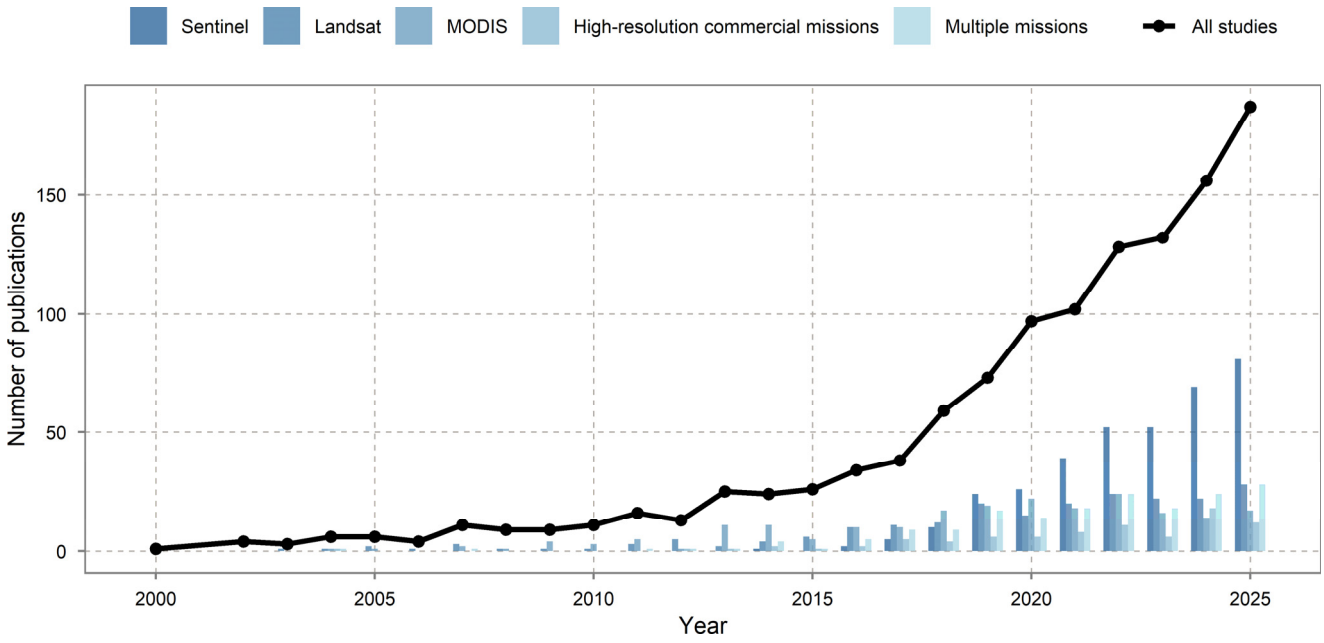


Figure 3. Number of publications per satellite missions utilized in crop yield prediction from the WoSCC database for the 2000–2025 period.

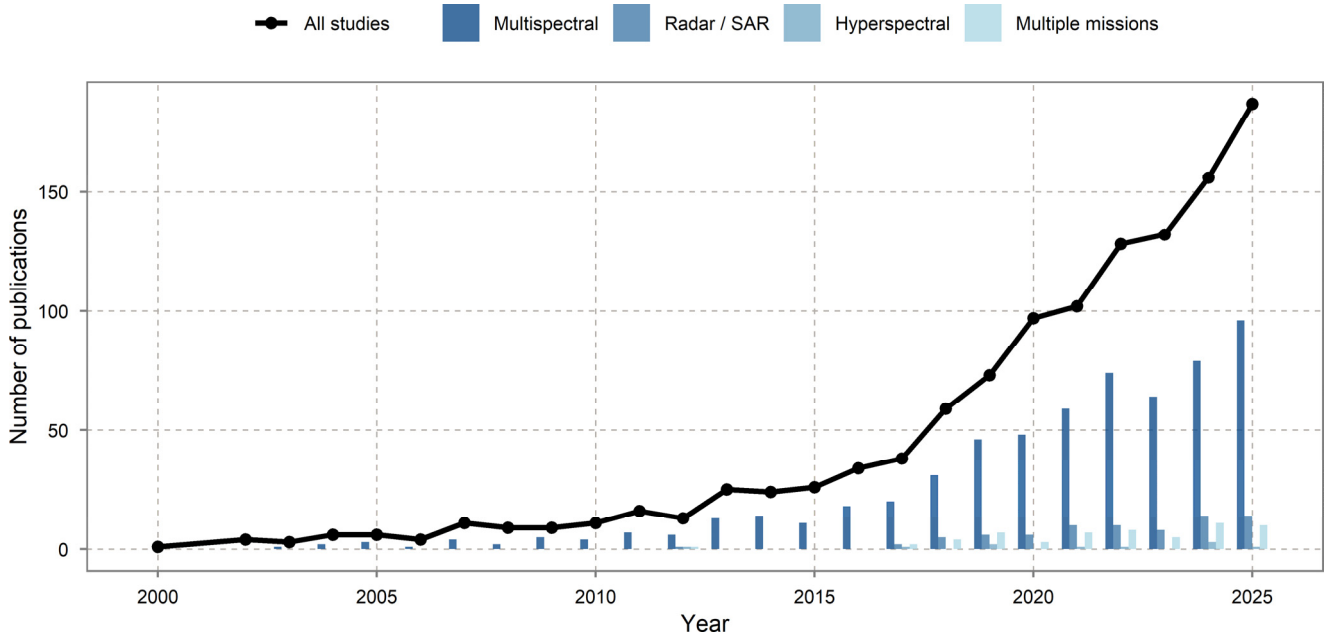


Figure 4. Number of publications per sensor type utilized in crop yield prediction based on satellite imagery from the WoSCC database for the 2000–2025 period.

3.3. Dominant Crops Used in Crop Yield Prediction Research

The scientific literature on satellite-based yield prediction is characterized by a high degree of crop-specific concentration, reflecting global food security priorities and economic significance. Figure 5 illustrates the distribution of research efforts across major crop types, revealing a clear hierarchy dominated by staple cereal grains. Wheat stands as the most extensively researched crop, with over 1500 studies identified in the WoSCC database. This dominance is driven by its status as a primary caloric source and its huge geographic footprint, which spans diverse agroecological zones from the big plains of the USA and Australia to the intensive production systems in China [61,62]. Maize follows closely as the

second most studied crop (~1000 studies), showing a particularly strong research density in the USA and China. The bibliometric data shows a high concentration of research in the USA and China, which together account for a significant share of global publications and citation centrality [63,64]. This aligns with these nations' roles as global leaders in maize production and their extensive use of satellite missions for monitoring. Rice, while ranking third (~450 studies), shows a distinct geographic specialization, with a heavy concentration of research originating from China and India, where rice serves as the cornerstone of national food security [63]. Soybeans represent the fourth pillar of research, frequently modeled alongside maize in the Americas due to their similar phenological cycles and rotational management [65]. This geographic pattern suggests a strong alignment between scientific output and national R&D investment levels in Earth observation infrastructure. While countries in South America and Africa are global agricultural leaders, their lower representation in the analyzed dataset could be a result of varying levels of access to sovereign geospatial infrastructure and differences in research funding cycles rather than a lack of agronomic importance [66,67]. However, beyond these "Big Four", there is a sharp decline in publication volume. Crops like cotton, barley, and rapeseed each account for fewer than 200 studies despite their significant industrial value. This indicates a research landscape that is heavily skewed toward large-scale, monocultural row crops [68]. Such crops are spectrally and structurally more conducive to observation by moderate-resolution sensors like Landsat and Sentinel-2, which have historically defined the field [69,70].

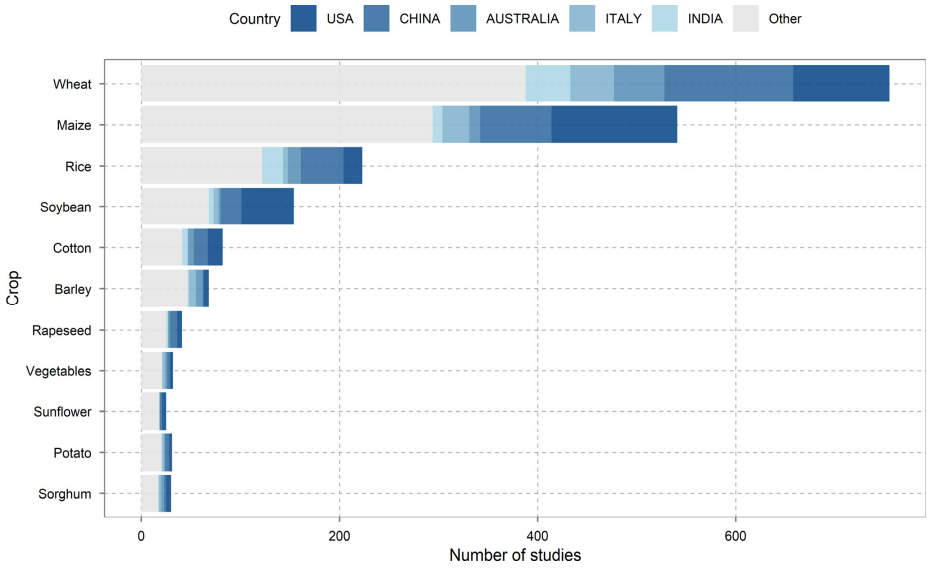


Figure 5. Number of publications per dominant crop type in crop yield prediction based on satellite imagery from the WoSCC database for the 2000–2025 period.

3.4. Specialty and Minor Crops: Current and Future Research Perspectives

Specialty crops, including vegetables, fruits, and industrial crops like cotton, remain significantly underrepresented in the remote sensing literature compared to major grains. The inherent challenges in predicting yields for these crops often stem from high subfield heterogeneity and complex canopy architectures that do not follow the uniform growth patterns of traditional cereal "row crops" [71]. Furthermore, many specialty crops have shorter, more intensive growing cycles or staggered harvest dates, which necessitate higher temporal and spatial resolutions than those typically required for large-scale wheat or maize monitoring [72,73]. While recent studies have begun to address these gaps by applying ensemble machine learning for processing tomatoes or sweet corn, the relatively low publication counts for crops like sorghum, sunflower, and diverse vegetable classes

(Figure 5) highlight a persistent "global caloric security bias". Research efforts have historically gravitated toward crops that dominate global trade and human energy intake, often overlooking the digital monitoring needs of complex horticultural or regional systems [74–76]. A critical synthesis of performance metrics across the literature shows that while staple cereals (wheat, maize) achieve high prediction accuracy ($R^2 > 0.80$), specialty crops often peak at R^2 values between 0.60 and 0.70 due to non-linear growth patterns and higher subfield variability. Addressing these limitations is crucial for expanding the benefits of precision agriculture to a broader range of agricultural stakeholders [73,77].

The distribution of the scientific literature reflects a profound concentration on staple crops that underpin global caloric intake and international trade. Figure 5 illustrates the absolute dominance of cereal grains, while Figure 6 reveals the accelerating temporal trajectory and scientific impact of these research efforts over the last two decades. Figure 6 demonstrates a clear inflection point around 2015–2016, where the volume of publications (bubble size) and scientific impact (color intensity) for all major crops increased exponentially. This period coincides with the launch of the Sentinel-2 mission, which provided the high-frequency, red-edge-rich data necessary for precise phenological tracking [78]. While wheat and maize consistently maintain the highest average citations per article (indicated by darker blue shades in historical years), the recent 2020–2025 period shows a massive "bubble" of productivity, indicating that the field is currently in its most active expansion phase. As shown in Figure 6, research for these crops only began to stabilize after 2010. The challenges associated with high subfield heterogeneity, plastic mulching in vegetable production, and varied harvest cycles have historically limited satellite-based yield modeling [74,79]. However, the rising frequency of publications for these minor crops in the 2023–2025 window suggests that the community is successfully transitioning from staple-only models to more diverse and inclusive agricultural monitoring frameworks [80–82].

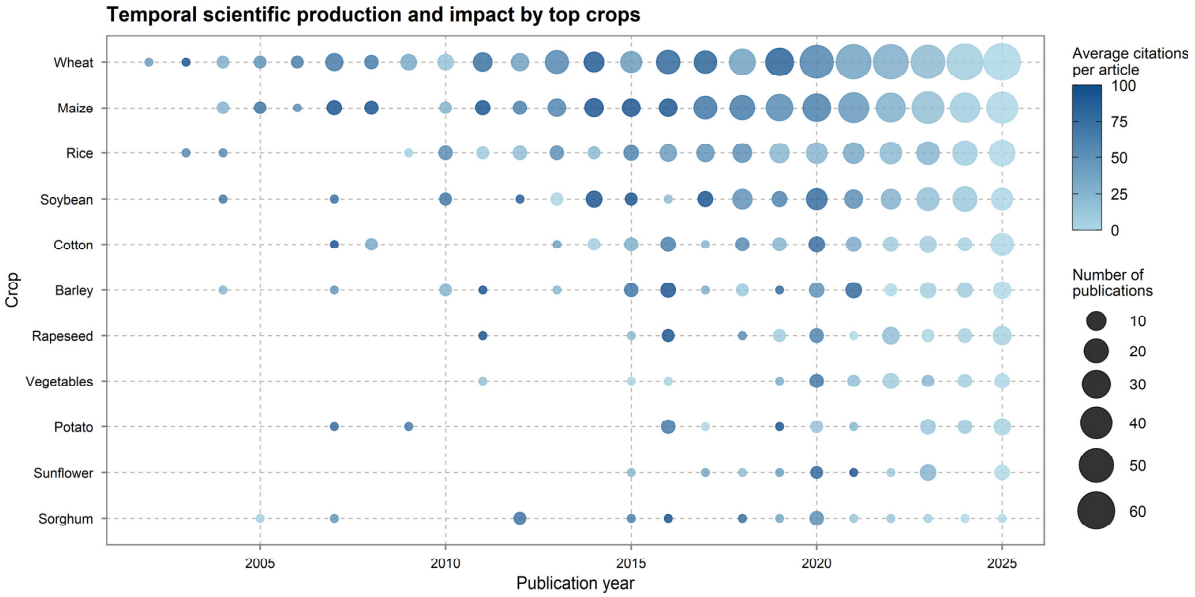


Figure 6. Temporal distribution of indexed publications per dominant crop type in crop yield prediction based on satellite imagery from the WoSCC database for the 2000–2025 period.

3.5. Evolution of Dominant Trends in Research Topics

The thematic trajectory of satellite-based crop yield prediction research reveals a clear transition from fundamental spectral observations to sophisticated, data-driven modeling frameworks. This section analyzes the shifts in research focus by examining keyword

dynamics and the emergence of specialized sub-topics over the 2000–2025 period (Figure 7). The evolution reflects a move from early empirical relationships based on simple vegetation indices toward the current dominance of machine learning, deep learning, and multi-sensor data fusion. These trends underscore the increasing complexity of the field and the ongoing search for models that balance high-dimensional predictive accuracy with biophysical interpretability [83–86]. The thematic evolution of the field began between 2005 and 2010 with foundational spectral observations, where yield and vegetation indices served as primary motor themes supported by basic remote sensing and machine learning. During this initial phase, research relied heavily on empirical relationships derived from MODIS and Landsat data [78]. The subsequent 2010–2015 period marked the emergence of the Sentinel era, characterized by mission-specific research where Sentinel-2 and NDVI models solidified their roles as motor themes, while Sentinel-1 appeared as a specialized niche for radar-based monitoring [87]. Between 2015 and 2020, methodological consolidation occurred as machine learning gained centrality for handling high-dimensional time series, while leaf-area index and winter-wheat research branched into niche specializations. The current 2020–2025 landscape is defined by a revolution in deep learning and Sentinel-2-based prediction, shifting the discipline from simple regressions toward complex architectures capable of modeling non-linear biophysical processes [88–90]. As precision agriculture matures into a central theme, artificial intelligence and data fusion emerge as the next frontiers to bridge multi-source sensing with operational field applications. Overall, this trajectory illustrates a successful migration from basic vegetation monitoring to integrated, AI-driven frameworks, with Sentinel-2 serving as the consistent technological anchor.

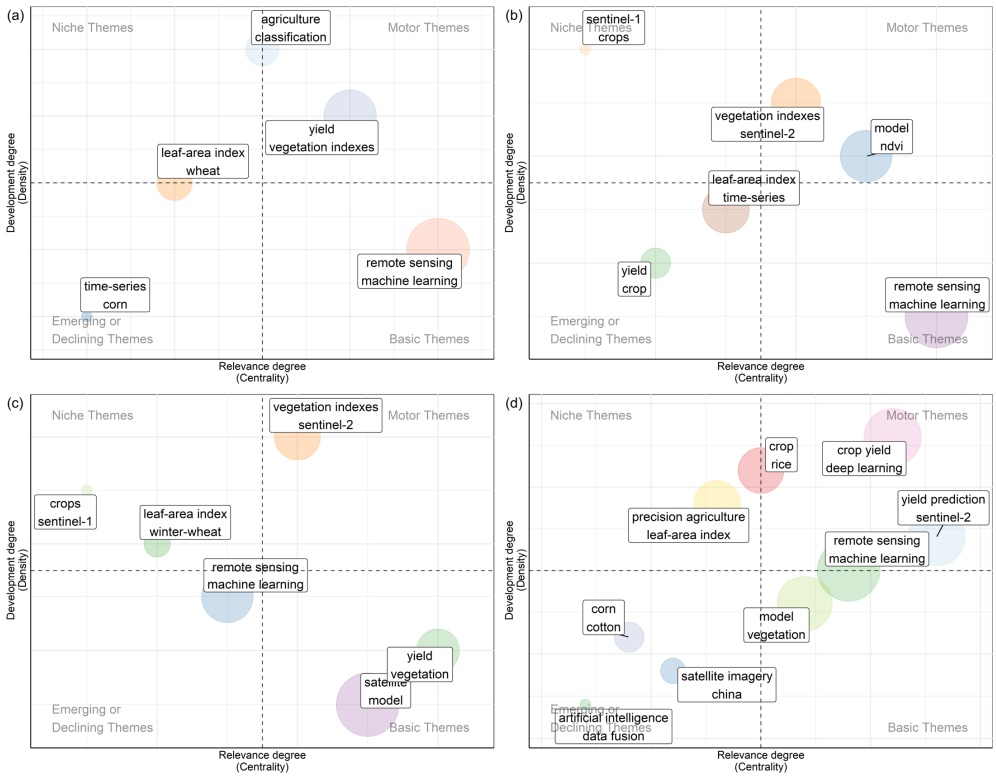


Figure 7. Visualization of trend dynamics for dominant themes in crop yield prediction based on satellite imagery for: (**a**) 2005–2010, (**b**) 2010–2015, (**c**) 2015–2020 and (**d**) 2020–2025.

The keyword co-occurrence network in Figure 8 provides a comprehensive mapping of the conceptual structure within the satellite-based crop yield prediction field, revealing four distinct thematic clusters that define the research landscape. The size of the nodes represents the frequency of keyword appearances, while the thickness of the edges indicates

the strength of their co-occurrence, illustrating a highly interconnected and multidisciplinary domain. The conceptual structure of the field, as mapped through the keyword co-occurrence network, reveals a highly interconnected and multidisciplinary domain where the blue cluster serves as the primary biophysical and modeling foundation. This methodological core is anchored by central keywords such as leaf-area index, vegetation indexes, and model, which maintain strong linkages to satellite-derived reflectance data. Furthermore, the proximity of soil-moisture to this grouping underscores a growing scientific imperative to integrate subsurface water status into comprehensive biophysical growth models to improve predictive accuracy [61,76,91].

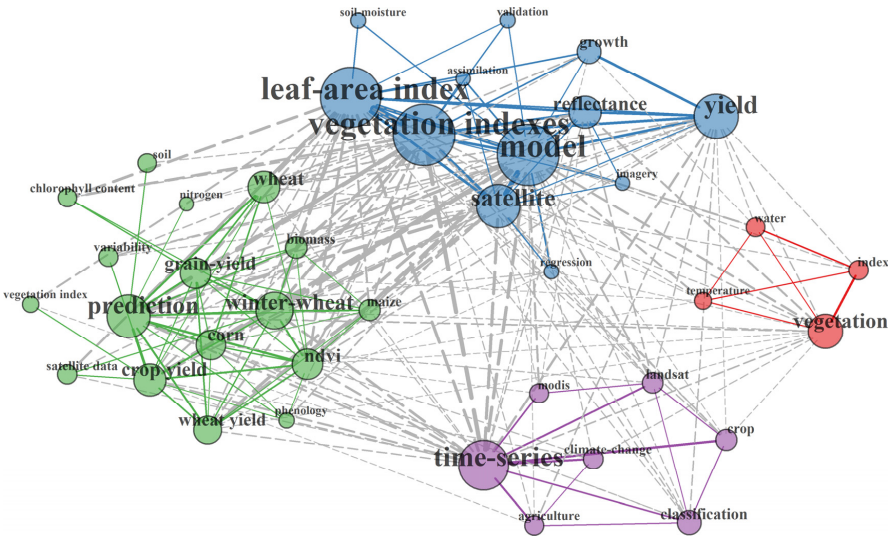


Figure 8. Keyword co-occurrences in publications on crop yield prediction based on satellite imagery from the WoSCC database for the 2000–2025 period.

The operational dimension of the research is represented by the green cluster, which dominates the predictive analytics landscape. This thematic area focuses on specific crop types, most notably wheat, winter-wheat, and corn, exploring their intricate relationships with grain-yield and the NDVI. The robust connection observed between prediction and variability within this cluster highlights a concerted effort to model and account for the interannual yield fluctuations driven by environmental stressors [92,93]. Simultaneously, the purple cluster emphasizes the data infrastructure supporting these efforts, where time-series analysis serves as a critical bridge between the Landsat and MODIS missions. This shift toward dense multitemporal analysis for classification and the tracking of climate-change impacts signifies the research community's heavy reliance on long-term historical archives to enhance the robustness of modern models.

The biophysical and environmental drivers are further articulated in the red cluster, which, although the smallest, is highly integrated. By linking vegetation and index with temperature and water, this cluster confirms the essential role of meteorological covariates in refining spectral-based yield estimations, particularly regarding evapotranspiration and drought monitoring [89,90]. Ultimately, the network architecture demonstrates that while the field is segmented into specialized sub-domains, the terms satellite, yield, and model act as the universal gravitational centers of the research. The strong link between the NDVI and the broader vegetation indexes node confirms that despite the emergence of advanced machine learning techniques, classical spectral proxies remain the primary bridge between raw satellite data and actionable yield predictions [70].

4. Global Distribution of Crop Yield Prediction Research Based on Satellite Remote Sensing

The data on the H-index for top countries based on total papers and citations from the WoSCC database for the 2000–2025 period in Figure 9 suggest a high concentration of research output in several countries, with the USA and China having dominant leads in total publications and the H-index. The leading position of the USA is indicative of the long history of the country investing into Earth observation programs, including Landsat and MODIS missions [94], as well as integration of agricultural research institutions with data infrastructure. The large number of publications in China indicates a rapid development of research potential and growing interest in the problem of food security and precision agriculture (Figure 10). The average rate of citation is lower than that of the United States, but the high H-index demonstrates the presence of a slowly increasing amount of impactful works, especially in the fields of yield prediction on a large scale and application of machine learning methods [95–97]. All eight leading countries in total publications also produced their record number of publications in the past two years (2024 and 2025), suggesting that crop yield prediction research based on satellite imagery will be of further interest globally. In addition to lacking in total publications due to lower populations than leading countries, Greece, the Netherlands and Canada were ranked below the USA in terms of average citations per publication. Generally, Africa and South America were underrepresented in rankings of top countries regarding total publications in crop yield prediction based on satellite imagery, even though the areas are susceptible to food insecurity and climate changes. This geographical gap indicates a significant gap in research and highlights the necessity to include more developing areas in future research. Models trained on more extensively researched crops and areas could be used for prediction in data-scarce contexts using transfer learning, which could narrow this gap. Open benchmarking datasets could also help create standard evaluation comparisons and help accelerate global advances in methods.

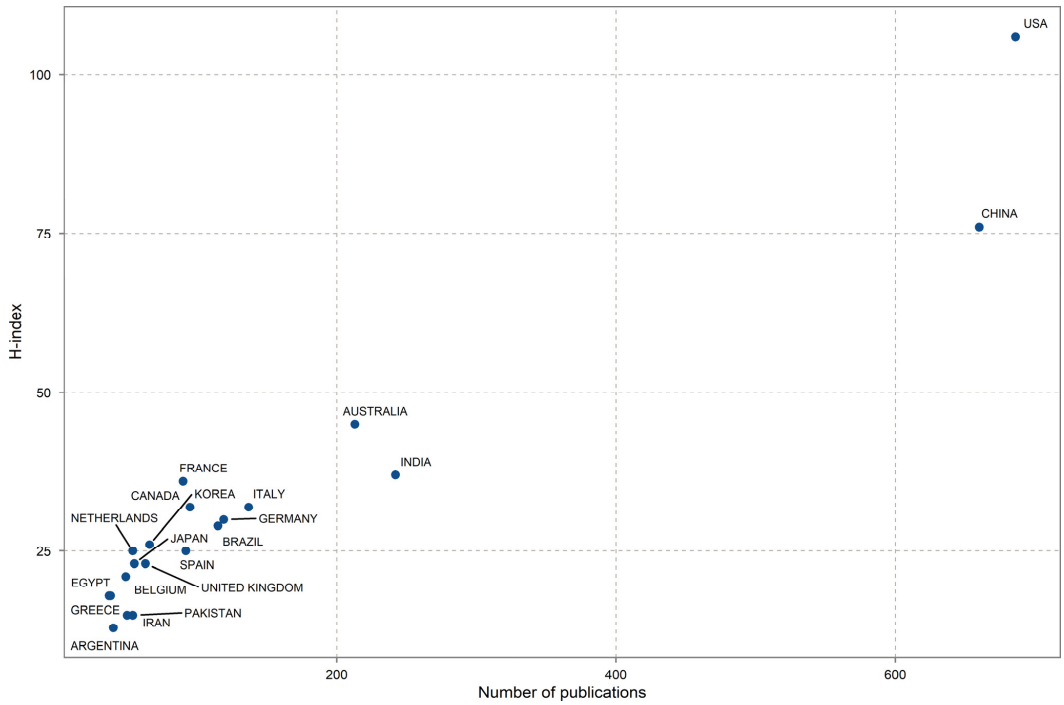


Figure 9. Scatterplot representing country H-index values and total publications related to crop yield prediction based on satellite imagery from the WoSCC database for the 2000–2025 period.

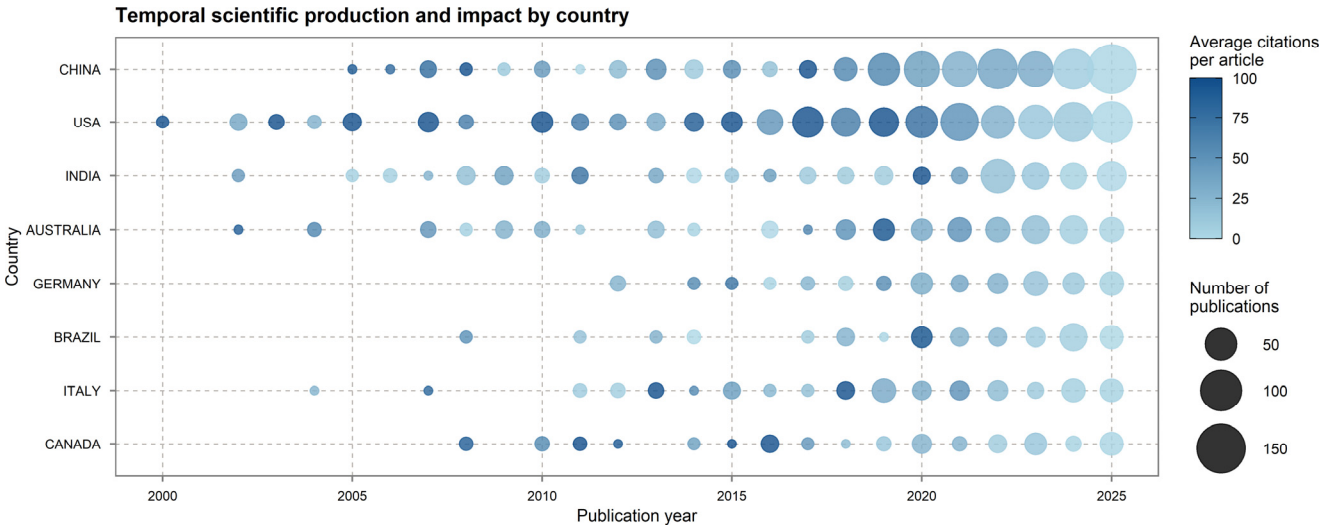


Figure 10. Temporal distribution of indexed publications for dominant countries in crop yield prediction based on satellite imagery from the WoSCC database for the 2000–2025 period.

5. Emerging Technologies in Crop Yield Prediction Based on Satellite Remote Sensing

The classical approaches of crop yield prediction that essentially relied on field observations, historical yield records, and empirical relationships were usually limited by spatial coverage, time resolution, and scalability. New technologies allow more time- and location-specific prediction of crop productivity as a result of combining high-resolution satellite data with machine learning algorithms and process-based modeling methods [18,98]. These novelties continue to produce new possibilities for crop monitoring, yield prediction and data-based agricultural decision-making. This especially includes interpretable machine and deep learning, integration of land surface phenology in yield prediction, multi-source data fusion approaches and crop growth simulation models.

5.1. Interpretable Machine Learning in Satellite-Based Yield Prediction

Although complex machine learning and deep learning models frequently enable high yield prediction accuracy when used on multitemporal satellite imagery, they have the disadvantage of being black-box models [99], which disables interpretability of affecting environmental impacts and limits practical implementation in the agricultural decision-making process. Satellite imagery provides an extensive range of covariates in crop yield prediction models [98], such as vegetation indices, spectral bands, radar backscatter measurements and phenology-related indicators. The interpretable machine learning provides insights into the non-linear relationship between covariates and crop yield, and since these approaches are model-agnostic [99], they have been successfully utilized with tree-based methods [100] and neural networks [101]. Permutation feature importance, partial dependence plots, accumulated local effects and SHAP (SHapley Additive exPlanations) values were frequently utilized methods for interpretability, quantifying and visualizing the importance of satellite-observed covariates [102]. It is especially useful when incorporating metrics of land surface phenology, since interpretability tools can determine which periods of the year are main drivers of crop yield prediction. Also, interpretability improves interregional model transferability and robustness, as interpretable covariates reduce the chances of false correlations and overfitting to particular data. However, the robustness of interpretability approaches is highly dependent on the validation design, and the reliability of model interpretability is increased when the spatiotemporal validation design is clear, such as leave-one-year-out cross-validation or spatial holdout [21].

5.2. Deep Learning and Process-Based Crop Models

Deep learning is another state-of-the-art processing approach, especially for large and multidimensional datasets that are frequently produced by the current satellite imagery time series [103]. Convolutional neural networks (CNNs) can identify the spatial features in the imagery, including canopy structure and field heterogeneity that most traditional machine learning algorithms fail to capture [104]. In the meantime, recurrent neural networks (RNNs) and long short-term memory (LSTM) networks are utilized to represent temporal dependences, with the result that the phenological progression throughout the growing season can be incorporated into the predictive models [105]. A significant advantage of deep learning is that they can directly utilize raw satellite imagery, which may eliminate the use of its derivatives, such as vegetation indices. Deep learning models can detect subtle signs of crop stress, biomass growth, or water shortage that can occur before a yield can be observed to be reduced by learning hierarchical representations of spectral and spatial data [106]. Deep learning methods have also shown high predictive ability when used with ensemble models or hybrid modeling frameworks in both field-scale and regional yield prediction [107]. Metrics for model evaluation dominantly include the root mean square error (RMSE) and the coefficient of determination (R^2), which are used to quantify absolute prediction errors and to explain variance [108]. Concerning the design of accuracy assessment approaches, most studies relied on random hold-out splits and k-fold cross validation, with the presence of validation methods that are temporally and spatially defined (leave-one-year-out, spatial cross-validation) due to their ability to more specifically evaluate model generalization [109].

Mechanistic (or process-based) crop models explicitly model the physiological and biophysical mechanisms that control crop growth and development, providing crop yield prediction based on weather conditions, soil properties, crop genotypes, and management practices, which offers a clear and theory-based model in interpreting the yield formation [110]. Models like DSSAT, APSIM, WOFOST and AquaCrop have been used to describe main processes such as photosynthesis, biomass growth, phenology, uptake of water and nutrients and adaptation to stress [111,112]. One of the key advantages of the process-based models is a high level of physical and biological basis, so they can be used to extrapolate across years, areas, and climatic conditions, including situations that are not adequately covered by historical data [113]. Furthermore, their direct description of phenology makes yield predictions directly related to developmental stages, management, and environmental limitations. Nevertheless, conventional process models tend to be restricted with large data needs and imprecision in the parameters [114]. Predictable simulations require precise inputs including soil specifications, cultivar-specific parameters and management information, which are often imprecise or geographically dissimilar at regional or global levels. These have historically limited their operational use in the prediction of large areas [115].

5.3. Land Surface Phenology as a Yield Predictor

The land surface phenology is important for producing temporal information regarding the crop growth dynamics, which are utilized as covariates for crop yield prediction using machine learning [116]. Based on time series of satellite-based vegetation indices or reflectance, land surface phenology metrics are used to measure significant growth periods [117,118]. The combination of these phenological indicators into predictive models allows a process-based interpretation of crop development, allowing the association of environmental conditions with observed productivity phenotypes [17]. Phenology-dependent methods are especially useful in highly variable areas with large interannual errors, where fixed associations between spectral indices and yield cannot describe environmental effects like drought, heat stress, or late planting [119]. Land surface phenology

integration enhances the model robustness, temporal interpretability, and the sensitivity of yield prediction models to crop management practices by aligning yield prediction models to the periods of critical development stages [17]. Also, the use of land surface phenology with machine and deep learning models enables the extraction of phenological signals in dense satellite time series automatically, enabling scalable and reproducible analysis of agricultural landscapes of large size [109].

5.4. Multi-Sensor and Data Fusion Approaches

Data fusion approaches that utilize complementary sensor capabilities have become possible due to the development of various satellite platforms. Multispectral sensors provide high-resolution spectral indices which are sensitive to vegetation greenness, as well as canopy chlorophyll content [46], while SAR sensors provide structural and moisture-related data, which is especially needed in the presence of clouds or in high-latitude areas [120]. Canopy temperature, evapotranspiration, and water stress estimates can be provided using thermal imagery on platforms, including MODIS [121]. Combining these different sources of data into one yield prediction system increases not only the accuracy but also the ability to overcome missing or noisy data. Such multi-sensor fusion is especially utilized using machine and deep learning methods, which can use heterogeneous input data and specialize in learning complex cross-sensor relationships [122]. This enables the combination of multispectral and radar data that enables the models to capture the biomass accumulation and canopy water status, both of which are robust predictors of the final crop yield [123]. Inclusion of advanced hyperspectral feature representation, which has potential for capturing complex spectral patterns, could also notably improve multi-sensor fusion as well as interpretable machine learning approaches for handling high-dimensional satellite data. In addition, data fusion approaches could also be used for early season yield prediction, when there are minor indicators of stress that can be observed by auxiliary sensors before harvesting [124]. The latest smart-driven developments in the Internet of Things (IoT) combine high-resolution satellite images with ground-based sensor networks, provide real-time monitoring of soil moisture, soil nutrients, and the microclimate, and incorporate the dynamics of crop growth [125]. Additionally, IoT-enabled systems improve the range and precision of predictive analytics as well as the flexibility of management options through the collection of big data and automated decision-making.

5.5. Study Limitations and Future Directions

Restriction to bibliometric analysis of studies indexed in the WoSCC in the English language omitted regional and national journals in non-English languages, as well as operational reports from international organizations, despite their practical significance. This might have produced geographical bias toward nations with a pronounced presence of WoSCC-indexed journals and research communities that focus on methodological innovation. For these reasons, the results interpreting geographical dominance and global research patterns should be considered as the relatively limited satellite-based yield prediction literature in the English language, as opposed to the complete international representation of research.

Despite numerous advances in satellite imagery and data processing for crop yield prediction, there are still a number of challenges facing the use of new methods of predicting yields. Regional, crop and season transferability are problems, as model performance is often determined by how well representative the training data is [126]. The quality and continuity of data, especially in cloud-prone or high-latitude regions, may be limiting factors to the usefulness of multispectral sensors and hence necessitate a strong multi-sensor approach. Moreover, the interpretability of complex machine and deep learning models

that can be used to act upon instead of making black-box predictions is still not widely accepted. In the future, new satellite missions, development of artificial intelligence and cloud computing are likely to contribute to even greater accuracy of the yield predictions, spatial resolution and timeliness of such predictions. The trends emerging are incorporation of hyperspectral imagery to detect nutrients and stress, real-time combination of multi-sensor data streams and designing of hybrid models combining physical knowledge with data-driven learning.

6. Conclusions and Future Considerations

The evolution of satellite-based crop yield prediction over the past quarter-century marks a definitive transition from basic spectral observations to an AI-driven infrastructure for global food security. Our systematic review of 1174 publications confirms that the era of static, linear regressions has concluded, replaced by a high-velocity period of multimodal data fusion and high-resolution monitoring democratized by missions such as Sentinel-2. However, our analysis indicates that while the field has reached a maturity plateau in multispectral modeling, future growth depends on addressing systemic constraints in model evaluation. The transition toward operational intelligence will be driven by the integration of digital twins and vision transformers, provided that issues of generalization and data missingness are systematically addressed through rigorous benchmarking. Ultimately, moving from reactive observation to proactive diagnostic tools requires a standardized approach for reporting uncertainty and model limits. The key insight derived from this review is that the shift toward multi-sensor fusion (optical + radar) and red-edge-rich data is the most effective strategy for overcoming the long-standing saturation problem of the NDVI, directly leading to more reliable subfield yield forecasting. A pivotal shift will involve moving beyond simple "canopy greenness" toward the diagnostic assessment of metabolic health, leveraging upcoming hyperspectral and thermal missions like CHIME and EMIT. By integrating these high-dimensional data streams into digital twin architectures, yield prediction will evolve from reactive observation into a proactive, immersive diagnostic tool for real-time crop stress management.

Ultimately, achieving geospatial equity through robust model transferability is not just a technical challenge; it is a global imperative. By 2030, our mission must evolve from simply observing the Earth to constructing a high-fidelity digital reflection of our ecosystems. This will ensure that every captured pixel directly contributes to securing the food supply of a rapidly changing planet, turning raw data into a sovereign asset for global resilience.

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